

The Ongiini-Eval-OW Benchmark: A Concept Paper for the Planned Benchmarking of Machine Translation and Large Language Models on Oshindonga and Oshikwanyama

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Abstract

Oshiwambo — a cluster of mutually intelligible Bantu languages spoken by over a million people across northern Namibia and southern Angola, and the home language of roughly half of Namibian households — has, to our knowledge, no published machine-translation evaluation benchmark. Major commercial services (Google Translate, DeepL, Microsoft Translator), open multilingual MT models (NLLB-200, MADLAD-400), and the open Masakhane checkpoint collection all lack coverage of either standardised dialect, Oshindonga or Oshikwanyama. We announce **Ongiini-Eval-OW**, a planned 600-item English $\leftrightarrow$ Oshindonga and English $\leftrightarrow$ Oshikwanyama benchmark with native-speaker references from two independent translators, a 30-item inter-translator agreement set, an 11-tag phenomenon-tagged stratification (≥ 30 items per tag), a deterministic 30 % blind split, and a reproducible scoring protocol over chrF++, BLEU, and COMET-22, supplemented by a 50-item human-evaluation round. We document the empirical coverage gap, the dataset composition, the launch-leaderboard model matrix across American, European, and Chinese frontier and open-weight systems, and the contribution pipeline. The dataset is targeted for first public release in Q4 2026; this v1.0 concept paper announces the design and the call for participation. Data and code will be released under CC-BY-4.0 and MIT respectively.

1 Introduction

Roughly one in two Namibian households uses an Oshiwambo dialect as their primary home lan-

*Reference translations for the Ongiini-Eval-OW dataset are by Kaarina Shoozi and Elizabeth Hamukwaya, independent translators based in Namibia. Both translators are credited as authors of the dataset in CITATION.cff; their substantive contribution is described in Section 3.6 and the Acknowledgments. Reference review and human evaluation are conducted with a volunteer community of ~ 40 Oshiwambo speakers engaged with the Ongiini AI project, acknowledged below.

guage (2011 Namibian Population and Housing Census; [Namibia Statistics Agency, 2024](#)). The cluster includes eight mutually intelligible Bantu varieties of which two — Oshindonga and Oshikwanyama — have standardised written forms, established orthographies, school curricula, and broadcast presence. Together they are spoken by approximately one to one and a half million people across northern Namibia and southern Angola, with Oshikwanyama particularly numerous in Angola’s Cunene Province. For a language community of this size, the absence of any public machine-translation evaluation benchmark is striking.

The absence is not theoretical. We have directly inspected the tokenizer of NLLB-200 ([NLLB Team et al., 2022](#)), the README and supported-language list of MADLAD-400 ([Kudugunta et al., 2023](#)), and the public Masakhane checkpoint repositories, and found no Oshiwambo coverage in any of them. We have additionally consulted the public language lists of Google Translate, DeepL, and Microsoft Translator, and found Oshiwambo absent from each. Frontier large language models — Claude, GPT, Gemini, Llama, Qwen — do produce Oshiwambo-like output when prompted, but the quality has not been measured by any benchmark and is therefore neither defensible nor falsifiable. Prior published parallel data for Oshiwambo amounts, to our knowledge, to the WON / “Writing Our Narratives” corpus ([Nekoto et al., 2022](#)) — a $\sim 5,500$ -sentence Oshindonga $\rightarrow$ English participatory training corpus that has been used for fine-tuning experiments but not as a standardised evaluation benchmark.

This paper announces **Ongiini-Eval-OW**, a planned 600-item evaluation benchmark for English $\leftrightarrow$ Oshindonga and English $\leftrightarrow$ Oshikwanyama machine translation and large-language-model output. The contribution is fourfold:

1. A reference-quality bilingual evaluation set.

600 English source items paired with native-speaker reference translations into both standardised dialects, contributed by two independent translators (Kaarina Shoozi and Elizabeth Hamukwaya). A 30-item subset is translated by both translators independently so that inter-translator agreement can be published as a variance measurement, rather than asserted as a uniform reference of uncertain quality.

2. **Phenomenon-tagged stratification with usable per-slice power.** Eleven phenomenon tags target known low-resource MT failure modes (negation (Hossain et al., 2020), noun-class agreement, code-switching, politeness register, polysemy, multi-sentence cohesion, and others), with each tag carrying at least 30 items so that per-slice scores in the 180-item blind split (30 %) carry roughly nine items per slice — a working minimum for slice-level comparison.
3. **Deployment-derived source items.** Roughly 30 % of the source items are paraphrased from production user queries to the Ongiini AI WhatsApp assistant, scrubbed of personally identifiable information (PII) and capped at three derived items per user. The result reflects the actual distribution of questions that Namibian users ask digital assistants in their language, not an encyclopaedic projection of it.
4. **A reproducible scoring protocol.** chrF++ (Popović, 2017), BLEU (Papineni et al., 2002), and COMET-22 (Rei et al., 2022) computed automatically over a published JSONL submission format, supplemented by a 50-item human-evaluation round with adequacy and fluency ratings and published Krippendorff’s α (Krippendorff, 2004).

This is a **concept paper for the planned v1.0 release**. An internal v0.1 build at 423 items exists and has been used to validate the pipeline end-to-end; the 600-item v1.0 build described in Section 3 is in production at the time of writing, with first public release of the dataset and an updated version of this paper targeted for Q4 2026. We publish this concept paper in advance to solicit model submissions, native-speaker reference review, and phenomenon-coverage contributions during the build phase — the patterns and the contact paths are detailed in Section 6.

2 Background: the Oshiwambo translation gap

2.1 The languages

“Oshiwambo” refers to a cluster of eight mutually intelligible Bantu dialects (Guthrie zone R.20; Maho, 2009 lists the cluster as R.21–24). Two of the eight have standardised written forms and dominate by written-text presence and L1 census share, as shown in Table 1.

Of the two, Oshikwanyama is the more numerous in both Namibia and Angola. At the 2011 Namibian Population and Housing Census, Oshikwanyama was the home language of 21.2 % of households and Oshindonga 15.1 %. The 2023 Namibian Census (Namibia Statistics Agency, 2024) reports the Aakwanyama (712,165 — 23.6 % of Namibians) as the largest ethnic group and the Aandonga (311,211 — 10.3 %) as the second largest. Ethnologue and Wikipedia cite roughly one million Kwanyama speakers in Cunene Province, Angola, as of 2024.

Both standardised dialects are agglutinating Bantu languages with rich morphology: active noun-class agreement chains, tonal phonology (not orthographically marked in everyday writing), and productive verbal derivation. Both are written in the Latin alphabet with a small set of regular orthographic conventions.

2.2 The coverage gap

We audited the major translation systems for Oshiwambo coverage in May 2026 and re-verified the findings in September 2026. For open systems we inspected tokenizer and repository artefacts directly; for closed/commercial systems we consulted the publicly published language list. The findings are summarised in Table 2.

Frontier LLMs do produce Oshiwambo output. Whether that output is accurate, fluent, register-appropriate, or substantively useful for real-world communication is an empirical question — and it is the empirical question this benchmark is designed to answer.

2.3 Why this matters

Roughly one in two Namibian households speaks an Oshiwambo dialect at home (49 %, 2011 census). Public services — health information, education, government communication, broadcasting — increasingly assume access to digital text. When the translation layer is absent, the entire

Dialect	ISO 639-1	ISO 639-3	Guthrie	Primary regions
Oshikwanyama	kj	kua	R.21	Ohangwena and Omusati regions (Namibia); Cunene Province (Angola)
Oshindonga	ng	ndo	R.22	Oshana and Oshikoto regions (Namibia)

Table 1: The two standardised Oshiwambo dialects covered by this benchmark.

System	Claims Oshiwambo	What we checked, and what we found
NLLB-200 (NLLB Team et al., 2022)	No	Inspected <code>special_tokens_map.json</code> on Hugging Face; no Oshiwambo language code present
MADLAD-400 (Kudugunta et al., 2023)	No	Inspected model README and supported-language list; no Oshiwambo language code present
Masakhane open checkpoints	No	Surveyed the public Masakhane MT repositories; no Oshiwambo checkpoint located
Google Translate (web)	No	Consulted public language list; Oshiwambo not offered as a translation option
DeepL	No	Consulted public language list; Oshiwambo not offered as a translation option
Microsoft Translator	No	Consulted public language list; Oshiwambo not offered as a translation option
Aya-23 (Aryabumi et al., 2024)	No	Cohere documents Aya-23 as covering 23 languages, none of which are African; Oshiwambo not among them
Meyabase Translate (https://www.meyabase.com/)	Yes (English ↔ Oshindonga)	Namibian NMT effort led by Axel Mukwena; ~ 70k-pair corpus per the project website; the only Oshiwambo-specific NMT system we are aware of
OpenAI / Anthropic / Google frontier LLMs	Best-effort, no formal claim	Produce plausible Oshiwambo output on prompt; quality un-benchmarked until now

Table 2: Empirical coverage audit (May 2026, re-verified September 2026). Inspection-based for open systems; documentation-based for closed APIs.

Oshiwambo-speaking population — over a million across Namibia and Angola — is structurally excluded from the digital infrastructure that other linguistic communities benefit from. This is not a niche research question; it is a digital-inclusion question. Joshi et al. (2020) classify Oshindonga as “Left-Behind” in their resource-tier taxonomy, the lowest of five tiers.

2.4 What changes when a benchmark exists

A benchmark provides (a) a measurable baseline against which any team’s MT or LLM advance can be cited; (b) an empirical pressure point on the vendors whose language-coverage decisions are otherwise opaque; (c) a scaffold for follow-on work on adjacent Namibian languages (Otjiherero, Khoekhoegowab, Rukwangali, Silozi) that today face the same coverage gap; and (d) a repeatable methodology — phenomenon-tagged, register-balanced, blind-split — that other low-resource language communities can adapt without re-deriving the design principles from scratch.

3 The Ongiini-Eval-OW dataset

This section describes the dataset as designed for first public release (v1.0). An internal 423-item

v0.1 build that exercises the same pipeline already exists; the 600-item v1.0 build described below is in production.

3.1 Composition

The v1.0 dataset is being built to six hundred (600) English source items, each paired with reference translations in both Oshindonga and Oshikwanyama, drawn from four provenance streams shown in Table 3.

The composition revision from v0.1 (423 items) was deliberate: the earlier dataset’s per-phenomenon slices were too thin to support meaningful per-slice statistical comparison in the blind split, and the chat/conversational register dominated more than was methodologically honest. v1.0 fixes both: every phenomenon carries 30+ items, formal/community/religious registers are bulked up, and the mined_paraphrased label replaces the misleading real_mined of the internal v0.1 schema.

3.2 Stratification

Items are stratified on three axes to allow per-slice scoring.

Length buckets (intentionally chosen to reverse v1’s short-phrase skew; longer items stress aggluti-

Source	Items	Description
v1_retained	150	Phrasebook-style items, curated down from a larger v1 set; longer constructions preferred over greetings.
mined_paraphrased	180	Inspired by production WhatsApp conversation logs: PII-scrubbed, then paraphrased into clean English while preserving register and intent. The published English source is not verbatim user text — the label is honest about this. Capped at three derived items per user to prevent dominance.
crafted	210	Phenomenon-tagged items authored to ensure each phenomenon slice carries at least 30 items (negation, code-switching, noun-class agreement, politeness register, etc.).
formal_drafted	60	Government, health, education, legal, and Namibian news/broadcast register items, drafted to anchor the benchmark in institutional language.
Total	600	

Table 3: Provenance composition of Ongini-Eval-OW v1.0.

nating morphology): 25 % short (1–6 words); 50 % medium (7–18 words); 25 % long (19+ words).

Domain mix (rebalanced from v0.1 to reduce chat skew): 38 % conversational chat; 22 % phenomenon-tagged challenge items; 22 % formal / institutional; 12 % community (family / village / community organising); 6 % religious.

Phenomenon tags are calibrated so each tag has ~ 9 items in the 180-item blind split, the threshold below which per-slice chrF++ comparisons become statistically uninformative. Items can carry multiple tags. Table 4 lists the eleven phenomena and item counts.

3.3 Splits

Every item will be run through every system being evaluated — the split is **not** about which items to test against, it is about which score is reported as the headline. The 600 items will be tagged into two subsets: a **development set** of 420 items (`in_blind_split = false`) usable for prompt engineering, error analysis, fine-tune training, or anything else that might involve looking at items and iterating in response; and a **blind set** of 180 items (30 %, `in_blind_split = true`) which submitters commit not to look at while building their system. The blind split is deterministic (`seed = 42`) and stratified by phenomenon $\times$ length $\times$ domain so each slice retains roughly its full-set proportion.

Both subsets will be translated and published at v1.0 release; you will be able to compute scores on either or on the full 600. The convention is summarised in Table 5.

Why hold a subset back at all, given the data is published? (i) *Prevent prompt-tuning leakage*. If everyone iterates against the full set, prompts implicitly fit those exact items and the headline score becomes inflated. (ii) *Slow training-data contam-*

ination. Once references are published they will be crawled and eventually appear in some model’s training corpus. At launch no system has seen the blind set because the dataset does not exist publicly yet; well-behaved teams undertake not to train on it later. (iii) *Match the standard MT-eval convention* used by FLORES (Goyal et al., 2022), WMT, and MAFAND-MT (Adelani et al., 2022), so reviewers and downstream users read the numbers as they expect to.

The blind split was sized at 30 % (rather than the conventional 20 %) so that each per-phenomenon slice in the blind set carries ~ 9 items — a working minimum for slice-level comparison, even if slice scores remain interpretive rather than headline (Section 4.5).

3.4 Schema

The TSV schema is summarised in Appendix B.

3.5 Safety and provenance

PII scrub at mining time. Phone numbers, ID numbers, addresses, and named individuals are stripped from source text before any human reviewer sees it. **Mined items are paraphrased, not copied.** The published English source is a paraphrase of the user input, preserving register and intent without leaking real-user text. This is reflected in the mined_paraphrased provenance label and is documented honestly rather than presented as raw “real-mined” data. **No machine-translated items shown to the human translator.** The translators work from English source only, with no automated proposal, to preserve reference independence. **Per-user cap of three items** prevents any single user’s speech patterns dominating the dataset.

Tag	Items	Tests
numbers_dates	40	Currency (N\$), dates, times, phone numbers, IDs
named_entities	30	Namibian places, ministries, common Namibian names
tense_aspect	30	Perfect / recent past / habitual (Bantu makes finer distinctions than English)
negation	30	Single, double, scope ambiguity — historically the #1 MT failure mode (Hossain et al., 2020)
code_switch	30	English loanwords embedded in Oshiwambo
pronoun_coreference	30	Ambiguous antecedents; Bantu noun-class pronouns force disambiguation
idiom_nonliteral	30	English idioms; translator matches local idiom or paraphrases
politeness_register	30	Tate / Meme / Kuku honorifics; elder / peer / child address
noun_class_agreement	30	Concord chains across subject prefix → verb → object marker → adjective
polysemy	30	Context-dependent lexical choice (“bank”, “right”, “school”)
multi_sentence	30	2–4 sentence mini-paragraphs testing discourse cohesion

Table 4: Phenomenon-tag inventory. Items can carry multiple tags.

Use	Which items?
Running a system to see what it outputs	All 600
Per-phenomenon / per-length / per-domain diagnostics	All 600
Tuning a prompt or fine-tuning a model	Dev set only
Headline leaderboard number, publishable claim	chrF++ on blind set

Table 5: Use-case convention for the development/blind split.

3.6 Inter-translator agreement

Kaarina Shoozi and Elizabeth Hamukwaya will both independently translate a designated **30-item overlap set per dialect** (`in_agreement_set = true`). Both translations will be published in `oshindonga_reference` (primary) and `oshindonga_reference_alt` (second translator), analogously for Oshikwanyama. From this overlap we will compute and publish: (a) character-level chrF++ self-similarity between the two translations of each item, reported as mean $\pm$ SD per dialect; (b) a qualitative disagreement count (lexical / morphological / pragmatic differences) coded by the dataset language coordinator; (c) a diff sample of representative disagreements in the methodology appendix.

We deliberately do **not** assert one translator’s reference is “correct” and the other “alternate”; both are valid references from fluent native speakers. The agreement set is a public window into the reference variance, not an adjudication exercise.

4 Benchmark protocol

This section specifies the experiment. External teams should be able to read this section and the submission specification at <https://github.com/sebkuepers/Ongiini/blob/main/data/>

oshiwambo_eval/submissions/README.md

and produce a valid leaderboard entry without contacting us.

4.1 Launch model matrix

To keep the launch claims honest, we separate *what we will run ourselves* from *what we invite external teams to submit* (Section 6). “Run ourselves” means we have working access today — either a paid API endpoint or a model that fits on our NVIDIA DGX Spark (128 GB unified memory, ~ 273 GB/s bandwidth). Anything else is welcome via the submission pipeline; we will not pre-commit to running it.

The launch leaderboard will cover at least one system per category and is geographically balanced across American, European, and Chinese state-of-the-art models.

Frontier proprietary LLMs (we run via API).

American: Claude Opus 5 (Anthropic, 2026); GPT-6 Astra (OpenAI, 2026); Gemini 3.1 Pro (Google DeepMind, 2026a). Chinese: DeepSeek V4.1 — both deepseek-chat (non-thinking) and deepseek-reasoner (thinking) (DeepSeek AI, 2026); Kimi K3 (Moonshot) (Moonshot AI, 2026); GLM-5.3 (Z.ai / Zhipu) (Z.ai (Zhipu AI), 2026). European: Mistral Large 3 (675B MoE, 41B active) (Mistral AI, 2025) and Mistral Medium 3.5 (128B dense) (Mistral AI, 2026a).

Open-weight LLMs (we run on Spark; quantised where needed). American: Gemma 4 26B MoE (3.8B active, comfortable fit on Spark) (Google DeepMind, 2026b); Llama 4 Scout (109B MoE / 17B active — fits at FP4 but slow at single-stream decode) (Touvron et al., 2025). Llama 4 is Meta’s last open-weight release; their frontier model is now closed (see Muse Spark callout below). European: Mistral Small 4 (24B dense,

Apache 2.0 — easy fit on Spark) (Mistral AI, 2026b). Chinese: Qwen 3.8 27B (Alibaba, Apache 2.0 — easy fit) (Yang et al., 2026); DeepSeek-R1-Distill-Llama-70B (quantised) for a smaller-footprint reasoning baseline (DeepSeek AI, 2026).

Running both API and on-Spark families lets us report whether sovereign on-device inference is viable for a Namibian deployment, not only what the largest cloud frontier model produces.

Dedicated machine-translation systems (open research models, we run on Spark). NLLB-200 (NLLB Team et al., 2022) and MADLAD-400 (Kudugunta et al., 2023) do not list Oshiwambo as a supported target, but both have been trained on multiple other Bantu languages from the same broad family — NLLB-200 includes Zulu (zul_Latn), Tswana (tsn_Latn), Xhosa (xho_Latn), Swahili (swh_Latn), Sotho (sot_Latn), Venda (ven_Latn), and others; MADLAD-400 covers 419 languages with similarly broad Bantu coverage. We will run each system with the closest-related-language target code (Tswana for the southwest Bantu zone) and report what comes out. This is explicitly a *zero-shot-transfer measurement, not a Oshiwambo translation score*: we expect low chrF++. The interesting datum is whether genuine Oshiwambo vocabulary or morphology leaks through, or whether the output collapses to the fallback language.

We exclude the consumer / commercial translation APIs (Google Translate, DeepL, Microsoft Translator) and Aya-23 (Aryabumi et al., 2024) from the launch matrix. None of them lists Oshiwambo as a target, and unlike the open research models above we cannot deliberately probe their zero-shot behaviour with a specific Bantu fallback code; they would either refuse or pick a fallback we don’t control. The coverage gap is already documented in Section 2.2; spending compute to confirm it twice does not add information.

Specialist systems (via external submission). Meyabase Translate (Axel Mukwena; English ↔ Oshindonga; <https://www.meyabase.com/>) — the only Oshiwambo-specific NMT system we are aware of, built on a ~ 70k-pair corpus and the closest peer effort to ours. We do not have access to run Meyabase ourselves; we expect the Meyabase team to run their system against the dataset and submit results via the submission pipeline (Section 6). Any Masakhane checkpoint that emerges with Oshiwambo coverage is welcomed via the same path.

Out of scope at launch. Models requiring

partnership-grade quotas, region-locked deployments we cannot access from Namibia, or on-premise hardware beyond a single DGX Spark are not pre-committed. They are warmly welcomed via the submission pipeline (Section 6) and will be added to the leaderboard with attribution as teams submit.

Meta Muse Spark

Meta’s closed-weight frontier model **Muse Spark** (Meta Superintelligence Labs, first released April 2026) deserves a specific call-out. In informal hand-testing of a small number of prompts through the meta.ai consumer interface, Muse Spark appeared to us to produce noticeably better Oshindonga and Oshikwanyama than the other systems we have informally probed, including several frontier LLMs. This is a striking and unexpected signal — a system with no documented Oshiwambo training claim looks qualitatively strong — but the observation is anecdotal: a handful of prompts, judged by us, with no metric and no held-out set. Measuring it properly is precisely what this benchmark is for.

At the time this benchmark was designed, Muse Spark had no public API, no on-premise option and no open weights, and the only way to interact with it was the consumer-facing meta.ai web interface — which made a reproducible 600-item run impossible. That objection has since been removed: with the release of **Muse Spark 1.1** in July 2026, Meta opened the Meta Model API to developers in public preview (Meta Superintelligence Labs, 2026), providing exactly the programmatic access the benchmark requires.

We therefore commit to evaluating Muse Spark 1.1 as part of the launch leaderboard.

At the time of writing we do not have API access — the public preview opened to United States developers first, with a waitlist for broader availability. Should access not be in place by first public release, Muse Spark moves to the submission pipeline (Section 6) on the same terms as every other system we cannot run ourselves, and we would welcome a submission from Meta directly. We flag the apparent capability here so that readers, partners, and Meta itself are aware that this dataset is ready to measure it.

4.2 Prompting protocol

To keep comparisons apples-to-apples: **LLMs** are evaluated with a single, public, zero-shot prompt template (Appendix C). No in-context examples —

this protects fairness across systems with uneven few-shot support and matches the regime in which an end-user would invoke the model. **MT systems** are called through native APIs with `target_lang` set to the dialect ISO code where supported, falling back to a related-Bantu code (Tswana) for systems that do not support Oshiwambo at all. **Pre/post-processing** is identical across systems: whitespace normalisation, line-by-line input, no surface form rewriting.

4.3 Automated metrics

chrF++ (Popović, 2017) — primary. Character-F-score with word-boundary weighting. Robust on character-rich, agglutinating Bantu morphology where token-level BLEU under-rewards near-misses. Well-established in FLORES (Goyal et al., 2022) and AfricaNLP work.

BLEU (Papineni et al., 2002) — secondary. Sentence-piece BLEU, included for legacy comparability with older MT literature and existing African-language work.

COMET-22 (Rei et al., 2022) — tertiary. Neural quality estimation with the English-side reference. *Reported with an explicit caveat:* COMET-22’s training data does not include Oshiwambo, so the score is suggestive, not definitive — interpret as a source-conditioned plausibility signal rather than a language-aware quality score. Wang et al. (2024) have recently extended COMET to African languages (AfriCOMET), which may apply once the model’s coverage list includes Oshindonga or Oshikwanyama; until then, COMET-22 is the reference.

All three will be computed by a public script (provided in the repository at <https://github.com/sebkuepers/Ongiini/tree/main/scripts>) that consumes the submitted model outputs in JSONL form plus the reference TSV. Outputs will be reported as overall scores plus per-slice matrices: per-phenomenon $\times$ per-length $\times$ per-domain $\times$ per-split.

4.4 Human evaluation

Automated metrics are necessary but not sufficient on low-resource languages. We pair them with a focused human-eval round on a 50-item stratified sample, mirroring the dataset’s overall phenomenon, length, and domain proportions.

Raters. Each item will be rated by approximately **5–10 independent native-speaker raters per dialect**, drawn from a volunteer community of

~ 40 Oshiwambo speakers already engaged with the Ongiini AI project. We are explicit about this recruitment source: the volunteer pool is self-selected toward supporters of the project, which is a real form of selection bias that a reviewer can correctly identify. We mitigate it through (a) blinding rater identity from system identity — raters see “Translation A” / “Translation B” and never know which model produced which (the standard MT-eval blinding convention); (b) randomising presentation order per item to neutralise position effects; (c) embedding 5–10 calibration items per session (known-good and known-bad translations) so that raters whose calibration scores are out of band can be filtered; (d) reporting rater demographics (dialect, region, age band, profession, regular-Ongiini-user status) in aggregate alongside the published scores; and (e) where logistically possible, supplementing with one or two external academic raters (Namibian university linguists or Masakhane-affiliated researchers) as a triangulation point. Individual rater identities are protected.

The trade-off is honest: this design exchanges one or two academic-credentialled raters for many more native-speaker raters drawn from the actual user community, with explicit disclosure and standard blinding controls. We believe this serves the methodological purpose of the human evaluation — detecting adequacy and fluency failures that automated metrics miss — better than a smaller academic-credentialled pool, while being transparent about where the raters come from.

Ratings. Two per item per rater: *Adequacy* (1–5, “does the translation preserve the meaning of the English source?”) and *Fluency* (1–5, “does the translation read naturally to a fluent speaker, independent of the source?”).

Inter-Annotator Agreement (IAA). Reported using Krippendorff’s α (Krippendorff, 2004) on the ordinal scale, computed both *within* the volunteer pool (volunteer-only α) and *across* the volunteer pool plus any external academic raters (cross-group α), so that the reader can compare the two and judge whether the volunteer pool’s variance differs substantively from the academic baseline. Items where ratings disagree by ≥ 2 points will be adjudicated by an external academic reader, with adjudication notes published in the leaderboard appendix.

Rater materials — rubric, anchor examples, reference cards (Appendix D) — will be published alongside the leaderboard so the human-eval

methodology is itself reproducible.

4.5 Reporting matrix and statistical power

The published leaderboard will be sliceable along these axes: per system $\times$ per dialect; per metric (chrF++, BLEU, COMET-22, human-adequacy mean, human-fluency mean); per slice (phenomenon $\times$ length bucket $\times$ domain $\times$ split).

Headline scores: chrF++ on the blind 180-item split. Everything else is interpretive. This convention is the publishable claim; slice-level scores are tools for understanding where each system breaks.

We are explicit about what the numbers do and do not support. **Headline blind-set scores** ($N = 180$) are sufficient for paired bootstrap confidence intervals at the per-system level. As a general guide from the chrF++ literature on similarly-sized test sets, differences smaller than ~ 3 chrF++ points are often statistically indistinguishable; we publish CIs explicitly so comparisons can be made honestly rather than relying on point estimates. **Per-phenomenon blind-set scores** (typically ~ 9 items per slice) are interpretive. They are useful for spotting *where* a system struggles (e.g. noun-class agreement, multi-sentence cohesion) but will not be reported as headline rankings. The leaderboard will present per-slice cells with explicit CIs and a footnote flagging small- N slices. **Per-system per-dialect** scores aggregate to the full blind set for both dialects (180 items each) and are reliable. **Item-level outputs** for every system will be published alongside the scores. Anyone wanting to run their own significance test — paired bootstrap, sign test, Wilcoxon — has the raw material.

This framing is conservative on purpose. We would rather under-claim on slice-level rankings than ship a per-phenomenon leaderboard cell that a reviewer can dismantle on power grounds.

5 Reproducibility

Dataset publication target. HuggingFace dataset at <https://huggingface.co/datasets/CommonIntelligenceFoundation/ongiini-oshiwambo-mt-eval>. The scaffolding (schema, README, license) matches the target. The dataset will be released under CC-BY-4.0 at first public publication.

Scripts. MIT-licensed at <https://github.com/sebkuepers/Ongiini/tree/main/scripts>. The pipeline

(`mine_eval_candidates.py`, `curate_mined_candidates.py`, `build_eval_v2.py`, `fill_baseline_translations.py`, `export_eval_set.py`) have been executed end-to-end on the internal v0.1 build and will be adapted for the v1.0 composition.

Baselines as reference. Pre-computed Claude Opus 4.7 and Gemma 4 26B outputs on the v0.1 build exist internally and validate the pipeline. Baselines will be re-computed against the v1.0 dataset (with Claude Opus 5 and Gemma 4 26B) before first public release and will form the first rows of the published leaderboard.

Citation. Citation File Format manifest at https://github.com/sebkuepers/Ongiini/blob/main/data/oshiwambo_eval/CITATION.cff renders as BibTeX, APA, and Zenodo metadata automatically. A Zenodo DOI will be minted at first formal publication.

Versioning. The first public dataset release will be pinned as v1.0. Schema-compatible updates will be minor (v1.1); schema-breaking changes will be major (v2). All releases will be archived on the dataset repository.

6 Call for participation and roadmap

Three contribution paths invite the wider community to populate the benchmark and strengthen the reference quality. Each has a defined pipeline so the cost-to-contribute is bounded.

6.1 Submit a model

The most-impact, lowest-friction path. Run your system against the 600 English source items, format the outputs as JSONL per the schema at https://github.com/sebkuepers/Ongiini/blob/main/data/oshiwambo_eval/submissions/schema.json, and open a pull request adding the submission under `data/oshiwambo_eval/submissions/<model-id>/` with a one-page model card. We validate the JSON, run chrF++, BLEU, and COMET-22 on every item, publish the per-slice matrices, and add the system to the leaderboard with attribution. Submissions received during the build phase (through Q4 2026) appear on the launch leaderboard as the first external rows alongside the Anthropic + Google baselines. For submissions received before the academic-paper cutoff (Q2 2027), the submitting team is offered co-authorship on the eventual

publication.

6.2 Review the reference translations

The dataset will ship with two translators and a 30-item inter-translator agreement set (Section 3.6). Broader native-speaker review beyond those two strengthens the reference further and surfaces dialectal variation that should itself be documented. Review a subset of the translations at your chosen size; submit alternates or flag disagreements via review forms linked on the dataset repository at first public release; optionally provide rater demographics (dialect, region) so aggregated rater context can appear in the published methodology. Accepted alternates will be integrated into a minor dataset version (v1.1 etc.) and reviewers credited in the Acknowledgments.

6.3 Propose phenomena and items

Phenomenon coverage in v1.0 is balanced for the constructions we know to test. Under-represented areas include Namibian-Afrikaans code-switching, proverbs, tone-affecting honorifics, and discourse markers. Propose new items with the required tags (length bucket, domain, phenomenon list) and proposed reference translations; optionally include a one-paragraph linguistic note explaining what the item tests. Accepted items will enter a future dataset version (v1.x for schema-compatible additions, v2 for schema changes); contributors are credited in the release notes.

6.4 Roadmap

The path from this concept paper to the first peer-reviewed publication is set out in Table 6.

6.5 Contact

GitHub issues, pull requests, and discussion at <https://github.com/sebkuepers/Ongiini> (the data/oshiwambo_eval/ directory). Email: sebastian@common-intelligence.org for partnership conversations, review requests, or submission questions. The Ongiini AI project: <https://ongiini.ai>.

7 Conclusion

We presented Ongiini-Eval-OW, a planned 600-item evaluation benchmark for English $\leftrightarrow$ Oshindonga and English $\leftrightarrow$ Oshikwanyama machine translation and large-language-model output, with native-speaker references from two independent translators, phenomenon-tagged stratification, an

inter-translator agreement set, and a reproducible scoring protocol over chrF++, BLEU, COMET-22, and a focused human evaluation round. The contribution is principally the dataset and its methodology — the empirical coverage gap (Section 2.2) and the design choices that follow from it (Sections 3–4). The dataset is in production at the time of writing and is targeted for first public release in Q4 2026; an updated version of this paper will be submitted as arXiv v2 alongside the release. We invite the wider community to populate the launch leaderboard, review the reference translations, and propose phenomenon-coverage extensions (Section 6).

Limitations

The benchmark is honest about its constraints. We surface them explicitly so reviewers do not need to.

- **Two translators, not three or more.** Kaa-rina Shoozi and Elizabeth Hamukwaya cover the full dataset, with a 30-item agreement-set overlap from which inter-translator agreement is reported (Section 3.6). This is stronger than the single-translator reference common in low-resource MT eval but weaker than a 3+ rater panel; the translation-review contribution path remains open as the explicit mitigation.
- **No back-translation validation.** A more rigorous protocol would back-translate references into English via a third translator blind to the source. Budget was directed toward broader phenomenon coverage and the inter-translator agreement set instead; future versions may add back-translation on a sampled basis.
- **Register bias.** Conversational chat remains the largest single domain at 38 % (rebalanced down from 47 % in v0.1). Long-form discourse, literary, and scientific registers are under-represented; the benchmark is for sentence-level MT and short-paragraph cohesion only.
- **$N = 600$ is mid-size.** Compared to FLORES-200 (NLLB Team et al., 2022) (3,001 per language) or MAFAND-MT (Adelani et al., 2022) (5,000+ per pair), Ongiini-Eval-OW is smaller. We deliberately optimised for density of useful items in the deployment register

When	Milestone
September 2026	This concept paper (v1.0) submitted to arXiv as cs.CL. Email outreach to external prospective submitters (Meyabase, Masakhane-affiliated teams) with the arXiv link, inviting submissions for the launch leaderboard. The 423-item internal v0.1 build has been fully translated into both dialects by both reference translators, validating the translation pipeline end-to-end at production scale.
Q4 2026	600-item English source set locked: new crafted and formal items authored, second Spark mining run executed and curated, pipeline scripts updated for the v1.0 schema. Translator handoff packets generated (one document per translator with their assigned items in randomised order), including the 30-item agreement-set overlap. Volunteer-pool onboarding for human evaluation begins in parallel. Claude Opus 5 and Gemma 4 26B baselines computed against the locked English source (independent of translator timeline).
End of 2026	Translator deliveries imported, references finalised, inter-translator agreement computed and published. Dataset v1.0 goes live on HuggingFace under CC-BY-4.0. First public leaderboard launches with the two Anthropic + Google baselines plus any external submissions received during the build phase.
Q1 2027	Volunteer-pool human-evaluation round 1 (50-item sample, 5–10 raters per dialect) completes. Krippendorff’s α computed within-volunteer and cross-group (with any external academic raters who participate). Concept paper v2 submitted to arXiv with finalised numbers and the live HuggingFace URL.
Q2 2027	Academic paper submission. Target venues (in order of preference): AfricaNLP Workshop (co-located with ACL or ICLR); LoResMT at EMNLP; ACL Findings; LREC. Submission cycle is bound by conference deadlines and runs after the artefact release.

Table 6: Roadmap from concept paper to first peer-reviewed paper.

and phenomenon coverage over raw N ; the contribution pipeline is the path to growth.

- **No audio.** The benchmark is text-only. Pronunciation, tone, and prosody are not directly tested. This is a real gap for Oshiwambo, which is a tonal oral vernacular as much as a written language.
- **English-source bias.** Dialect $\leftrightarrow$ dialect translation is out of scope. Third-language pivots (e.g. Oshiwambo $\leftrightarrow$ Portuguese, for Angolan speakers) are not tested.
- **No long discourse.** The longest items are 19+-word multi-sentence paragraphs; document-level coherence is not benchmarked.
- **Adjacent Namibian languages are explicitly out of scope.** Otjiherero (hz / her), Khoekhoegowab (naq — covering the Nama and Damara dialect continuum), Rukwangali (kwn), and Silozi (loz) face the same coverage gap; they are future work but **not part of any commitment this benchmark makes**.
- **Pre-release status.** This is a concept paper for the planned v1.0 release; an internal v0.1 build at 423 items exists and has been used to validate the pipeline, but the full 600-item v1.0 composition is in production at the time of writing. Final numbers (inter-translator agreement, recomputed Claude Opus 5 and Gemma

4 26B baselines, populated leaderboard) will appear in the arXiv v2 of this paper to be submitted at first public dataset release (Q4 2026).

Anticipated reviewer critiques and our responses

We expect peer reviewers to raise the following objections; we surface them here rather than wait for them in review.

- “ $N = 600$ is small for an MT benchmark.” We agree. Our defence is that every phenomenon slice carries ≥ 30 items and the blind split is 30 % to keep per-slice power meaningful; that the contribution pipeline is designed for growth; and that for the specific deployment we serve (a free WhatsApp AI assistant in Namibia) benchmark *register* matters more than benchmark *volume*.
- “*Mined items are paraphrased, so they’re not really deployment-derived.*” We agree; the provenance label `mined_paraphrased` is honest about this. They are inspired by real-traffic distributions, not raw user text. They remain the closest publishable proxy to the actual deployment surface available, because verbatim user text cannot be released for privacy reasons.
- “*Per-phenomenon scores on ~ 9 -item blind slices are noisy.*” We agree, and we report

them with explicit confidence intervals and a footnote flagging small- N slices; we do not present per-slice scores as headline rankings (Section 4.5).

Ethics Statement

The Ongiini-Eval-OW benchmark is built on the following ethical posture, made explicit so the reader can verify it rather than trust it.

Consent and PII. The 180 source items derived from production WhatsApp queries to the Ongiini AI assistant (<https://ongiini.ai>) are *paraphrased*, not verbatim. Each source conversation is PII-scrubbed (phone numbers, ID numbers, addresses, personal names) before any human reviewer sees it, then rewritten into clean English by the dataset team preserving register and intent. The published English source is not attributable to any individual user, and verbatim user text is never released. Source users have agreed to the privacy policy at <https://ongiini.ai/privacy/>, which permits derived non-attributable use of aggregated signals for the explicit purpose of improving the assistant. A per-user cap of three derived items prevents any single user’s speech patterns dominating the dataset.

Translator labour and credit. Both reference translators (Kaarina Shoozi and Elizabeth Hamukwaya) are compensated for their work at market rates appropriate to professional Namibian translation work; their names appear in the Acknowledgments and in the dataset’s CITATION.cff with their consent.

Reference variance, not adjudication. The 30-item inter-translator agreement set publishes both translators’ work side-by-side. We do not assert one translator’s reference is “correct” and the other “alternate”; both are valid references from fluent native speakers, and the agreement set is a public window into the reference variance rather than an adjudication exercise.

Open licence as anti-extraction posture. The dataset is released under CC-BY-4.0, the code under MIT, and this paper under CC-BY 4.0. Anyone improving on Oshiwambo translation through use of the benchmark must credit the artefact and the named translators — a deliberate counterweight to the historical pattern of low-resource language work being absorbed into proprietary systems without attribution.

Rater recruitment disclosure. The human-evaluation raters are recruited from a volunteer

community of ~ 40 Oshiwambo speakers already engaged with the Ongiini AI project (Section 4.4). This is a self-selected pool, not a neutral sample. We disclose the recruitment source explicitly rather than present it as a conventional academic-recruit panel, and we mitigate the selection-bias risk through standard blinding of system identity, randomised presentation order, embedded calibration items, published rater demographics, and supplementary external academic raters where logistically possible. Volunteer labour is recognised in the Acknowledgments; volunteers who wish to be named individually have that option at their discretion.

Acknowledgments

The benchmark is built collaboratively. Specific acknowledgments will be expanded with each release.

Kaarina Shoozi and Elizabeth Hamukwaya — reference translators for both Oshindonga and Oshikwanyama. Both translators are credited as authors of the dataset in the CITATION.cff manifest.

The **Ongiini AI volunteer community** — approximately 40 Oshiwambo speakers from across northern Namibia who have engaged with the project as native-speaker reviewers, sample contributors, and human-evaluation raters. Their work makes the human-evaluation round of this benchmark possible. Per-release acknowledgments will expand with named contributors at each contributor’s discretion.

The Ongiini AI team at the Common Intelligence Foundation.

The authors and maintainers of the publicly available Oshiwambo reference materials we consulted while designing this benchmark, including *Hai ti! A Beginner’s Guide to Oshikwanyama* (Crane et al., 2004) and the Omniglot Oshiwambo phrasebook.

Meyabase (<https://www.meyabase.com/>, <https://github.com/meyabase>) — the Namibian Oshiwambo MT project led by Axel Mukwena. Meyabase’s $\sim 70,000$ -pair English $\leftrightarrow$ Oshindonga corpus and Neural Machine Translation tool is the closest peer effort to ours; we hope the Meyabase team will submit their system against this benchmark via the submission pipeline at first public release. We are grateful for their public pioneering work.

Nekoto et al. (2022) — the WON (“Writing Our Narratives”) corpus is the earliest published Os-

hindonga ↔ English parallel corpus we are aware of. Their participatory methodology and their honest documentation of the prior digitised data landscape (including that the only OPUS-listed Oshikwanyama corpus is in fact mislabelled German) directly influenced our approach.

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A Phenomenon tag definitions

The eleven phenomenon tags are intended to be empirically distinguishable failure modes for translation systems.

negation — sentences whose meaning hinges on a negation operator. Includes single (“I don’t have it”), double (“I never said I wouldn’t go”), and scope-ambiguous (“Not everyone came”) constructions. The historical #1 MT failure mode (Hossain et al., 2020).

numbers_dates — sentences containing numerals, monetary amounts (Namibian dollar, N\$), dates, times, phone numbers, or ID numbers. Tests the system’s ability to preserve numerical content verbatim while translating surrounding context.

named_entities — proper nouns: Namibian places (Windhoek, Oshakati), ministries, agencies,

and common Namibian personal names. Tests entity preservation across translation.

tense_aspect — Bantu languages make finer aspectual distinctions than English (perfect vs recent past vs habitual vs remote past). Sentences here test whether systems collapse the distinctions or preserve them.

code_switch — English loanwords embedded in Oshiwambo, or the reverse. Common in real WhatsApp register: “WhatsApp”, “ID”, “grant”, “Ministry” used in otherwise-Oshiwambo sentences.

pronoun_coreference — sentences with ambiguous antecedents that Bantu noun-class pronouns force the translator to disambiguate.

idiom_nonliteral — English idioms (“hit the books”, “raining cats and dogs”) whose literal translation produces a non-idiomatic Oshiwambo sentence.

politeness_register — honorifics (Tate / Meme / Kuku), elder/peer/child address forms, register-marking that has no direct surface-form equivalent in English.

noun_class_agreement — concord chains across subject prefix → verb → object marker → adjective. Bantu-specific; systems trained primarily on Indo-European data tend to fail here.

polysemy — English words whose Oshiwambo lexeme depends on context: “bank” (river vs financial); “right” (correct vs political); “school” (educational institution vs fish school).

multi_sentence — 2–4 sentence mini-paragraphs that test discourse cohesion across sentence boundaries (pronoun reference, tense consistency, topic chaining).

B Schema reference

Full schema for the public TSV at `data/oshiwambo_eval/data/eval_set.tsv`:
`id` (integer, 1..600, stable); `length_bucket` (enum: S / M / L); `domain` (enum: chat / formal / religious / community / challenge); `phenomenon_tags` (string, semicolon-separated subset of the 11 tags defined in Appendix A); `provenance` (enum: v1_retained / mined_paraphrased / crafted / formal_drafted); `english` (string, source sentence); `oshindonga_reference` (string, native-speaker reference, primary translator); `oshikwanyama_reference` (string, native-speaker reference, primary translator); `oshindonga_reference_alt`

(string, alternate reference from the second translator on the 30-item agreement set); oshikwanyama_reference_alt (string, alternate reference from the second translator on the 30-item agreement set); oshindonga_translator_notes (string, optional commentary); oshikwanyama_translator_notes (string, optional commentary); in_blind_split (boolean, 180 items / 30% marked true via stratified deterministic seed 42); in_agreement_set (boolean, 30 items per dialect with independent dual references).

Items will also be published as JSONL at data/oshiwambo_eval/data/eval_set.jsonl and as plain-text parallel files at data/oshiwambo_eval/data/{en,oshindonga,oshikwanyama},txt (one item per line; line N corresponds to id N). The blind and development subsets will be published as separate JSONL files (blind_split.jsonl, development_split.jsonl) for participants who want to operate strictly within one split.

C Prompting templates

LLM zero-shot template (used for all frontier and open-weight LLMs):

You are a professional translator translating English to {DIALECT} (an Oshiwambo language spoken in northern Namibia). Translate the following English sentence into natural, fluent {DIALECT}.

Only output the translation. Do not include explanations, glosses, or commentary.

English: {SOURCE}
{DIALECT}:

Substitutions: {DIALECT} is one of Oshindonga or Oshikwanyama; {SOURCE} is the English item. No system-prompt variation across LLMs; same template for all.

MT system templates — system-specific because of API heterogeneity. Documented per-system in the submission appendix at publication time.

D Human-eval rater rubric

Adequacy (1–5): “Does the translation preserve the meaning of the English source?”

- 5 All meaning preserved; no information lost or added.

- 4 Most meaning preserved; minor information missing or shifted.
- 3 Core meaning preserved; significant information missing or distorted.
- 2 Some meaning preserved; major elements wrong.
- 1 Meaning is wrong, missing, or unrelated to the source.

Fluency (1–5): “Does the translation read naturally to a fluent speaker?” (Rate independently of the source.)

- 5 Natural, native-sounding sentence.
- 4 Mostly natural; one or two minor awkward moments.
- 3 Understandable but noticeably non-native or awkward.
- 2 Difficult to understand; several errors.
- 1 Incomprehensible or ungrammatical.

Raters receive anchor examples for each score before the rating round. Adjudication by a third reader is invoked when raters disagree by ≥ 2 points on either dimension.

E Contributor code of conduct

Contributors to Ongiini-Eval-OW agree to:

- **Respect for the languages and their speakers.** Contributions treat Oshindonga and Oshikwanyama as the living, evolving languages of millions of speakers — not as resources to be extracted.
- **Attribution and consent.** Reference translations and reviewer contributions are credited at the contributor’s discretion; anonymous contribution is supported.
- **No machine-translated content in submissions.** Reference translations and reviewer alternates are native-speaker work, not MT post-edits.
- **Open licence.** All accepted contributions are released under CC-BY-4.0 (data) or MIT (code), consistent with the existing dataset and scripts.
- **Respectful engagement.** Discussions, code reviews, and dataset reviews are conducted with patience and care.